$$P(X=3) = \frac{e^{-\lambda}\lambda^3}{3!} = \frac{e^{-0.4}(0.4)^3}{3\times 2\times 1}$$

$$= \frac{0.6703\times 0.064}{6}$$

$$= 0.00715$$

**Example 10.15**

If $X$ is a Poisson variate with $P(X=0) = 0.2725$, find $P(X=1)$

***Solution:***

$X \sim P(\lambda)$

$\therefore P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}$ $\quad x = 0, 1, 2, \ldots\infty$

$P(X=0) = 0.2725$

$\frac{e^{-\lambda}\lambda^0}{0!} = 0.2725$

$e^{-\lambda} = 0.2725$

$\lambda = 1.3$ (from the table of values of $e^{-m}$)

$$P(x=1) = \frac{e^{-\lambda}\lambda^1}{1!} = \frac{e^{-1.3}\times 1.3^1}{1!}$$

$$= 0.2725\times 1.3$$

$$= 0.3543$$

**Example 10.16**

The probability of safety pin manufactured by a firm to be defective is 0.04. (i) Find the probability that a box containing 100 such pins has one defective pin . (ii) Among 200 such boxes, how many boxes will have no defective pin

***Solution:***

Let $X$ denote the number boxes with defective pins

$X \sim P(\lambda)$

$\therefore$ $P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}$ $\quad x = 0, 1, 2, \ldots\infty$

$p = 0.04$